Table 16: Fit Statistics for Unconstrained versus Constrained Models

| constrained | Df | AIC | BIC | Chisq | Chisq.diff | Df.diff | Pr..Chisq. |
|---|---|---|---|---|---|---|---|
| **LGBT** | | | | | | | |
| Unconstrained | 4 | 5738.17 | 5988.66 | 445.15 | | | |
| Constrained | 16 | 6205.33 | 6380.67 | 936.31 | 491.16 | 12 | 0 |
| **Abortion** | | | | | | | |
| Unconstrained | 4 | 8672.04 | 8922.54 | 448.95 | | | |
| Constrained | 16 | 9755.80 | 9931.15 | 1556.71 | 1107.76 | 12 | 0 |
| **Economic Attitudes** | | | | | | | |
| Unconstrained | 4 | 4636.64 | 4887.13 | 436.84 | | | |
| Constrained | 16 | 7310.98 | 7486.33 | 3135.18 | 2698.34 | 12 | 0 |
| **Party ID** | | | | | | | |
| Unconstrained | 4 | 6098.39 | 6348.88 | 662.96 | | | |
| Constrained | 16 | 9689.23 | 9864.58 | 4277.80 | 3614.84 | 12 | 0 |
| **Ideology** | | | | | | | |
| Unconstrained | 4 | 5217.96 | 5468.45 | 619.89 | | | |
| Constrained | 16 | 6746.99 | 6922.33 | 2172.92 | 1553.03 | 12 | 0 |

### 3.4 Crossed-lagged Coefficients from RI-CPLM models

Table 17: Cross-lagged coefficients from RI-CLPM models in GSS data

| dv | iv | est | se | pvalue |
|---|---|---|---|---|
| **Abortion Attitudes** | | | | |
| Politics | $Personality_{t-1}$ | 0.05 | 0.03 | 0.04 |
| Personality | $Politics_{t-1}$ | 0.07 | 0.06 | 0.13 |
| **LGBT Attitudes** | | | | |
| Politics | $Personality_{t-1}$ | 0.01 | 0.04 | 0.38 |
| Personality | $Politics_{t-1}$ | 0.05 | 0.04 | 0.13 |
| **Social Welfare Attitudes** | | | | |
| Politics | $Personality_{t-1}$ | 0.04 | 0.02 | 0.03 |
| Personality | $Politics_{t-1}$ | 0.01 | 0.03 | 0.32 |
| **Party ID** | | | | |
| Politics | $Personality_{t-1}$ | 0.01 | 0.02 | 0.37 |
| Personality | $Politics_{t-1}$ | 0.00 | 0.04 | 0.49 |
| **Ideology** | | | | |
| Politics | $Personality_{t-1}$ | -0.01 | 0.03 | 0.36 |
| Personality | $Politics_{t-1}$ | -0.03 | 0.03 | 0.18 |

## 4 CLPM and RI-CLPM

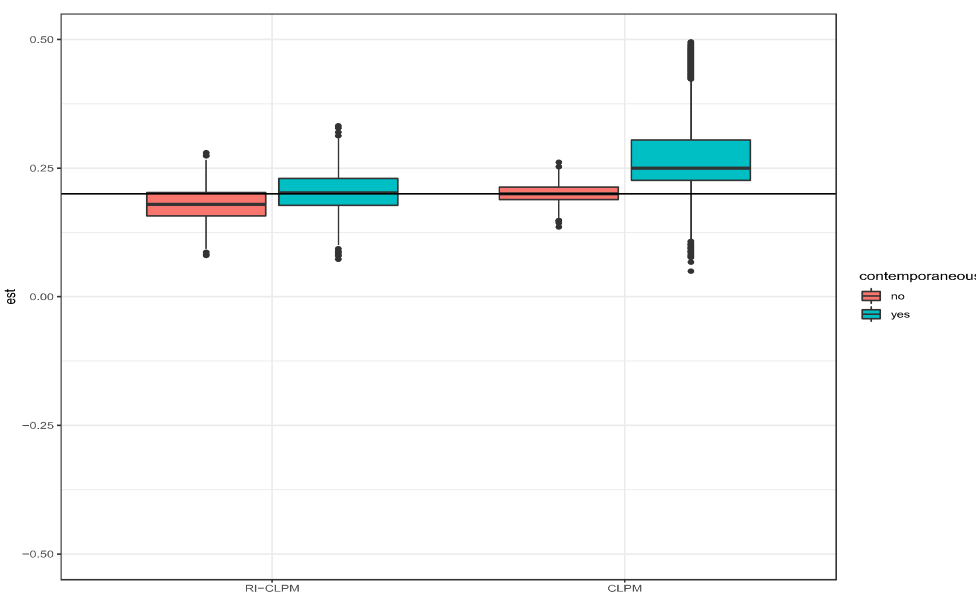
Though use of cross-lagged panel models [Finkel, 1995] is quite common and we presently pre-registered the plan to use them, it is important to note that this procedure has limitations. In particular, the traditional CLPM does not distinguish within-person and between-person variance, and is ultimately susceptible to omitted variable bias. Simply including individual-fixed effects can severely downwardly bias estimates, and even lead to sign reversals, when the number of waves of data is small [Nickell, 1981]. To ameliorate this issue, several models have been suggested [e.g., Allison et al., 2017, Hamaker et al., 2015] with the Random Intercept-Cross Lagged Panel Model (RI-CLPM) being perhaps the most popular.

The RI-CLPM decomposes observed scores into "grand means, stable between components and fluctuating within components" [Mulder and Hamaker, 2020, p. 2]. However, this approach also has serious drawbacks when the number of waves is small ($t<10$). First, "if the true causal effect of X on Y is contemporaneous rather than lagged, [these models] substantially underestimate the true coefficient size ..." [Leszczensky and Wolbring, 2019, p. 7]. For instance, if changing ones political attitudes affects ones reported personality in less than two years (the shortest time period between measurements in the three datasets we analyze), than the relationship between the variables will appear substantially smaller (possibly to the point of the direction flipping) than it actually is. Vaisey and Miles [2017, p. 62] argue that in 3-wave panel datasets that collect measurements infrequently (as the present ones do), such downwardly biased estimates "will likely be the rule rather than the exception." [Leszczensky et al., 2018]. While Vaisey and Miles [2017] focus on Allison's [2009] model, which is similar to the RI-CLPM, our own simulations demonstrate that a contemporaneous reciprocal relationship between two variables is severely biased when using the RI-CLPM.

We simulate two sets of three-wave longitudinal models. In the first, we explicitly set the contemporaneous relationship between the two sets of variables to zero. In the second, we allow for contemporaneous effects. We set the true relationship between lagged X and Y to =.2. The autocorrelation between lagged Y and Y was set to =.7, to approximately match the high stability of personality we see in our data (increasing autocorrelation or also setting the autocorrelation in the X variable to be high makes RI-CLPM results even worse). True correlations for all other variables are randomly set to lie between .1 and .3. We then simulate 4000 datasets: one for RI-CLPM with no contemporaneous effect; one for RI-CLPM with a contemporaneous effect; one for RI-CLPM with a contemporaneous effect randomized to be between .1 and .3; and one for CLPM with a contemporaneous effect randomized to be between .1 and .3. We plot the estimated coefficients between lagged X and Y in Figure 1.

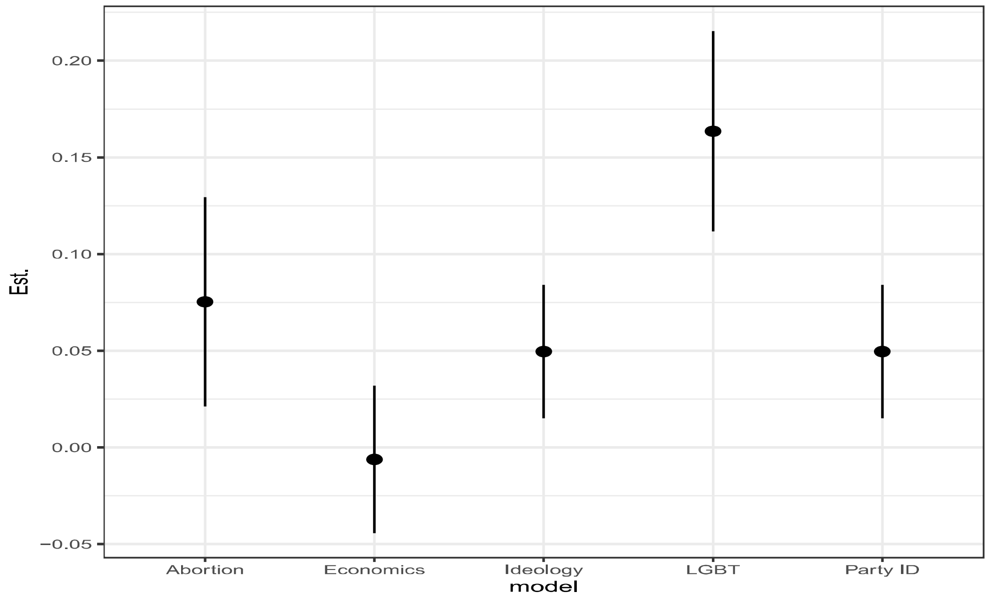
While RI-CLPM and CLPM models with no contemporaneous results recover, on average, the true relationship between lagged X and Y, the RI-CLPM model estimates are severely negatively biased when there is a contemporaneous effect. On average, the estimate is in the wrong direction. The CLPM model, on the hand, returns coefficients that are biased but far less so–the CLPM model is off by 50 percent while the RI-CLPM model is off by 100 percent.

Figure 1: Distribution of coefficients from model regressing lagged X on Y, wherein the true relationship is .2




Allison [2015, p.1] suggests a test for this bias: "estimate models that include both contemporaneous and lagged predictors. If a one-year lag is the correct specification, then the contemporaneous effect should be small and not statistically significant. If, on the other hand, the contemporaneous effect is large and significant, it should raise serious doubts about validity of the method and the kinds of conclusions that can be drawn." That positive and significant contemporaneous relationships between the outcome measure and the predictor appear frequently in our data raises serious concern about the validity of this method with the present data. To illustrate this, we plot the contemporaneous relationship between authoritarianism and political attitudes from the GSS in Figure 2 where four of the five contemporaneous effects are positive.

Figure 2: Contemporaneous relationship from cross-lagged models regressing authoritarianism on political attitudes




Another concern is statistical power. The RI-CLPM model typically has less statistical power than does the traditional cross-lagged panel model, especially when comparing variables that are relatively stable [Allison, 2009, Hill et al., 2020]. This increases the possibility of Type II error. Underpowered studies have a lower likelihood of replicating [Asendorpf et al., 2013].

Given the here discussed limitations of the RI-CLPM, we believe the traditional cross-lagged panel model estimates are more credible than the RI-CLPM models. Regardless, as our discussion above illustrates, drawing causal conclusions from longitudinal survey data is a challenging and fraught endeavor. We presently report panel data results using the traditional, and preregistered, CLPM and include the RI-CLPM models in this appendix (Netherlands: SI 1.6; Germany: SI 2.7; United States SI 3.7). The RI-CLPM analyses suggest an absence of causal influences between personality and politics in either direction.